CONCEPTUAL SERIES
Kafka Streams

Name: .... Cohort/Batch: .... Date: .... Score: ....

KAFKA STREAMS PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Data Engineering

TOTAL: 10 QUESTIONS

Q1 What is Kafka Streams and what problem in Data Engineering does it solve? *Threat Model*

Q2 What are the core components or concepts in Kafka Streams? *Architecture*

Q3 How does Kafka Streams compare to its main alternatives? *Configuration*

Q4 How do you get started with Kafka Streams for a new project? *Hardening*

Q5 What configuration options most affect Kafka Streams's behavior in production? *SSO / IdP*

Q6 How does Kafka Streams handle reliability and high availability? *Observability*

Q7 What observability does Kafka Streams provide out of the box? *Lifecycle*

Q8 What are common performance bottlenecks in Kafka Streams? *Compliance*

Q9 What security best practices apply when running Kafka Streams? *Testing*

Q10 What mistakes do teams commonly make when adopting Kafka Streams? *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Kafka Streams is a tool or — Mitigation

Kafka Streams is a tool or platform that automates or simplifies a specific concern in the Data Engineering domain, reducing manual effort and operational complexity for engineering teams.

A2 Kafka Streams has key building blocks — Primitives

Kafka Streams has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A3 Kafka Streams trades off simplicity against — Hardening

Kafka Streams trades off simplicity against feature richness differently than its alternatives. Choose Kafka Streams when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A4 Install Kafka Streams via its package — Gotchas

Install Kafka Streams via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A5 Production-critical settings typically include — Federation

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A6 Kafka Streams provides HA through leader — Observability

Kafka Streams provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A7 Kafka Streams exposes metrics (Prometheus — Automation

Kafka Streams exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A8 Performance bottlenecks typically stem from — Governance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A9 Run Kafka Streams with the principle — Assessment

Run Kafka Streams with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A10 Common pitfalls include skipping the — Optimization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.